



ML BASED ON EMAIL SPAM DETECTION

P R E S E N T A T I O N 2 0 2 6



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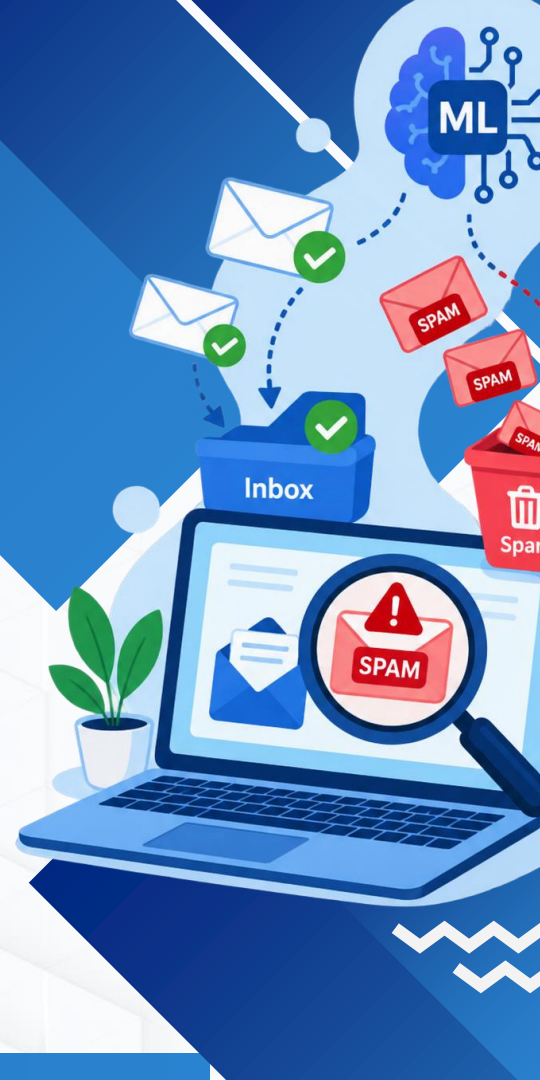


Guide

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OUTLINE

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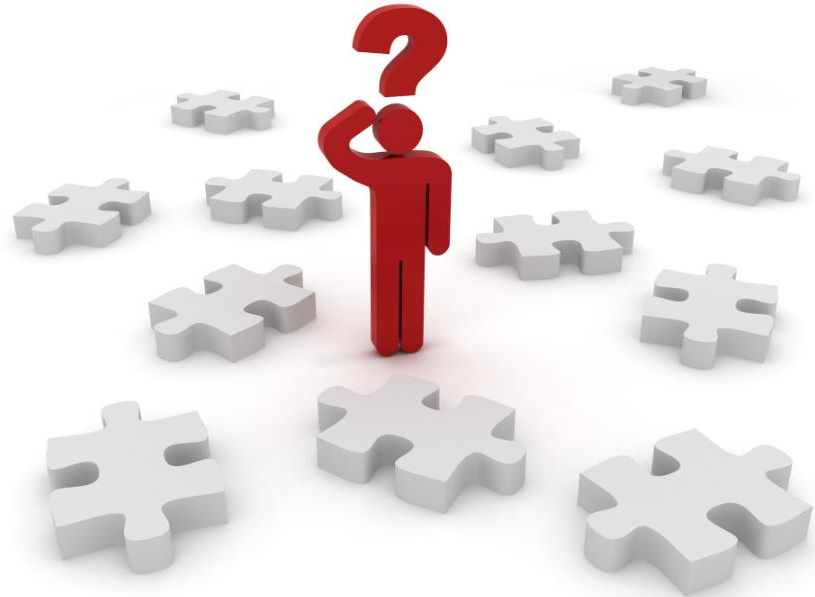
Abstract

Email spam detection is an important application of machine learning that helps identify and filter unwanted or harmful emails. The proposed system uses machine learning algorithms to analyze email content and classify messages as **spam or legitimate (ham)**. Text preprocessing techniques such as removing unnecessary characters, tokenization, and feature extraction are used to convert email messages into useful data. The trained machine learning model learns patterns from previously labeled emails and predicts the category of new incoming messages. This system helps reduce unwanted emails, improve email security, and provide users with a safer and more efficient email experience.

Problem Statement

Problem Statement

The increasing number of spam emails creates problems such as **unwanted messages, phishing attacks, fraud, malware, and loss of time**. Traditional email filters may fail to identify new and changing spam patterns. Therefore, there is a need for an **ML-based email spam detection system** that can automatically analyze email content and accurately classify messages as **spam or legitimate (ham)**.



Proposed Solution

The proposed solution uses machine learning to automatically detect and classify emails as ****Spam or Not Spam****. Email data is collected and preprocessed, followed by feature extraction from the email content. A machine learning model is trained using these features and used to classify new emails. The system is evaluated using accuracy and other performance measures to improve spam detection.



System Architecture

Data collection

Collect emails from users or a dataset for analysis.

Data preprocessing

Clean email text and extract useful features using TF-IDF or similar methods.

Machine learning model

Train a model such as Naive Bayes, SVM, or Logistic Regression

Spam classification

Classify each email as **Spam** or **Not Spam**.

Live Demo of Project

Here's the list of some demo of the projects -



email input&preprocessi ng

Enter the email and clean
the text for analysis.



Feature extraction & ml prediction

Convert the email
into features and
predict using the
trained ML model.

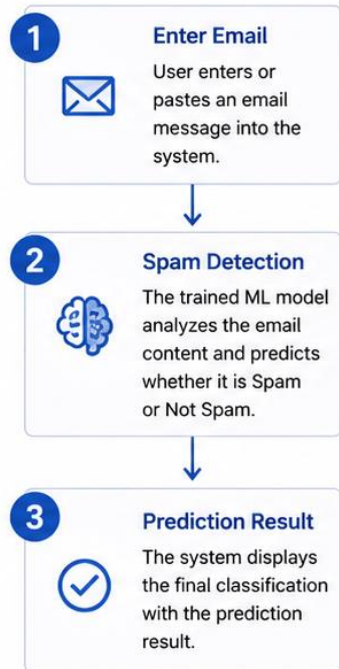


Spam result

Display whether the
email is classified as
spam or legitimate.

Live Demo of ML-Based Email Spam Detection

Machine Learning model classifies emails as Spam or Not Spam



 **Email Spam Detection**Home Predict About

Check if your email is Spam or Not Spam

Enter Email Content

Subject: Limited time offer! Claim your free voucher now!!!

Dear User,

Congratulations! You have been selected to receive a \$1000 gift card. Click on the link below to claim your reward now. This offer is valid for the next 24 hours only.

Click here to claim: <http://freereward.com/claim>

Thank you,
Best Offers

Predict Clear

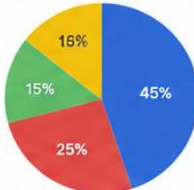
Prediction Result

 **Spam**

Confidence Score
98.67 %

Email Analysis Details

Total Characters	: 345
Total Words	: 62
Total Links	: 1
Model Used	: Multinomial Naive Bayes
Vectorizer	: TF-IDF



Indicator	Percentage
Promotional Words	45%
Links	25%
Special Characters	15%
Others	15%

Top Indicators

-  Promotional Words
-  Links
-  Special Characters
-  Others



Performance Evaluation Metrics (ML-Based Email Spam Detection)



These metrics are used to evaluate how well our machine learning model identifies spam emails and classifies them correctly.

1 Accuracy



Measures the percentage of emails correctly classified as Spam or Not Spam.

Higher Accuracy = Better Performance

Formula

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Where,

TP = Spam correctly detected
TN = Non-spam correctly detected
FP = Non-spam incorrectly marked as spam
FN = Spam incorrectly classified as non-spam

Interpretation



Higher accuracy indicates that the model correctly classifies more emails, both spam and non-spam.

2 Precision



Measures how many emails predicted as Spam are actually Spam.

Higher Precision = Fewer legitimate emails marked as spam

Formula

$$\text{Precision} = \frac{TP}{TP + FP}$$

Where,

TP = Spam correctly detected
TN = Non-spam correctly detected
FP = Non-spam incorrectly marked as spam
FN = Spam incorrectly classified as non-spam

Interpretation



Higher precision means most of the emails predicted as spam are actually spam, reducing false alarms.

3 Recall



Measures how many actual Spam emails are correctly detected.

Higher Recall = Fewer spam emails missed

Formula

$$\text{Recall} = \frac{TP}{TP + FN}$$

Where,

TP = Spam correctly detected
TN = Non-spam correctly detected
FP = Non-spam incorrectly marked as spam
FN = Spam incorrectly classified as non-spam

Interpretation



Higher recall means the model detects most of the spam emails, reducing the number of spam emails that are missed (false negatives).

4 F1-Score



Combines Precision and Recall into a single score.

Higher F1-Score = Better balance between Precision and Recall

Formula

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Where,

TP = Spam correctly detected
TN = Non-spam correctly detected
FP = Non-spam incorrectly marked as spam
FN = Spam incorrectly classified as non-spam

Interpretation



Higher F1-Score indicates a better balance between precision and recall, giving a more reliable overall performance.



Why These Metrics?

These metrics help us understand how accurately and reliably our machine learning model identifies spam emails while reducing both missed spam and incorrectly blocked legitimate emails.



Smarter Emails
A Safer Inbox

Conclusion

The ML-based Email Spam Detection system provides an effective way to identify unwanted and harmful emails automatically. By preprocessing email content, extracting important features, and applying a machine learning model, the system classifies emails as spam or not spam. This helps users reduce unwanted messages, improve email security, and save time.



Future Scope

In the future, the ML-based Email Spam Detection system can be improved by using advanced machine learning and deep learning models to achieve higher accuracy. It can be enhanced to detect new types of spam, phishing emails, and malicious links in real time. The system can also learn from user feedback and continuously update the model to handle changing spam patterns.



Thank You